Environmental Data Science: Homework #10
Explainability
Due: Before Class on 11/11/26

Permutation Feature Importance

What are the most important variables for the prediction tasks we've done in class thus far?

Revisit the following two learning tasks, and plot the variable permutation feature importances as a barchart using the optimal models you used in the prior homeworks:

- EPA AQS Data linear classification
- NOAA Buoy Predictions for XGBoost, decision trees, and a linear method. Are they the same or different for different models?

Methods Description

Read Flora et al. (2024) A Machine Learning Explainability Tutorial for Atmospheric Sciences (https://doi.org/10.1175/AIES-D-23-0018.1):
https://journals.ametsoc.org/view/journals/aies/3/1/AIES-D-23-0018.1.xml

1. How do they describe the permutation feature importance method we learned in class?
2. Describe an additional method for explainability from this paper. How does it work? Is it model agnostic or model specific?
3. Find potential scikitlearn functions for 5 of the methods described in the paper. Note: You don't have to implement them.